LLT Harm & Risk Analysis Traceability Document

Analysis Title: Seattle LLT Dataset Harm and Risk Keyword Tagging

Analyst: STLCA Data Analyst

Date of Analysis: January 25, 2026

Analysis Environment: Windows 10/11, Python 3.x, pandas 3.0.0, venv_stlca

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1. Data Sources & Provenance

File Name	**Rows**	**Source**	**Date Accessed**	**Notes**
District_1_LLT_ALL_ENRICHED.csv	7,527	C:\Users\stlca\stlca_data\data_raw\	2026-01-25	Successfully processed
District_2_LLT_ALL_ENRICHED.csv	8,873	C:\Users\stlca\stlca_data\data_raw\	2026-01-25	Successfully processed
District_3_LLT_ALL_ENRICHED.csv	11,092	C:\Users\stlca\stlca_data\data_raw\	2026-01-25	Successfully processed
District_4_LLT_ALL_ENRICHED.csv	6,676	C:\Users\stlca\stlca_data\data_raw\	2026-01-25	Successfully processed

District_5_LLT_ALL_ENRICHED.csv	7,232	C:\Users\stlca\stlca_data\data_raw\	2026-01-25	Successfully processed
District_6_LLT_ALL_ENRICHED.csv	6,295	C:\Users\stlca\stlca_data\data_raw\	2026-01-25	Successfully processed
District_7_LLT_ALL_ENRICHED.csv	7,895	C:\Users\stlca\stlca_data\data_raw\	2026-01-25	cp1252 encoding required
CITYWIDE_ALL_20251231_ENRICHED.csv	233,543	C:\Users\stlca\stlca_data\data_raw\	2026-01-25	Successfully processed

Output Files: data_clean\outputs\[filename]_HARM_AND_RISK.csv (8 total files)

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2. Analysis Methodology

Processing Pipeline

```text
1. Read CSV → dtype=str, encoding="cp1252", low_memory=False
2. Keyword match on Description column → 3 boolean flags:
   - HealthSafetyHabitability
   - HarassmentRetaliationDisplacement
   - RightsViolations
3. Write tagged CSV → encoding="utf-8"
4. Aggregate summary → groupby("District")
```

Script Location

text

C:\Users\stlca\stlca_data\scripts\RUNME_LLT_ALL_HARM_AND_RISK_data_raw.py

Execution Command

powershell

```
cd C:\Users\stlca\stlca_data
.\.venv_stlca\Scripts\activate
C:\Users\stlca\stlca_data\scripts\stlca_py.cmd -u
C:\Users\stlca\stlca_data\scripts\RUNME_LLT_ALL_HARM_AND_RISK_data_raw.py
```

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3. Replication Guide

Prerequisites

1. Python 3.8+ with pandas 3.0.0+
2. Virtual environment: `C:\Users\stlca\stlca_data\.venv_stlca`
3. Input files in `data_raw\` directory
4. Batch launcher: `RUN_LLT_HARM_AND_RISK.bat`

Step-by-Step

text

```
1. Double-click RUN_LLT_HARM_AND_RISK.bat (or run command above)
2. Script reads 8 input CSVs
3. Creates 8 output CSVs in data_clean\outputs\
4. Prints per-district + citywide summary to console
5. Window pauses - review output, close when ready
```

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4. Analysis Results Summary

Per District Counts (True flags only):

text

```
District 1: health=8786, harassment=2617, rights=2238
```

District 2: health=10250, harassment=2539, rights=2591
District 3: health=9755, harassment=3738, rights=3426
District 4: health=6816, harassment=2434, rights=1890
District 5: health=7102, harassment=2602, rights=2537
District 6: health=6051, harassment=3006, rights=1895
District 7: health=6125, harassment=2352, rights=2629
nan: health=240, harassment=120, rights=107

Citywide Totals: 55,125 health/safety + 19,408 harassment + 17,313 rights violations

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5. Threats to Validity

Threat	**Description**	**Impact**	**Mitigation**
Encoding Issues	District 7 required cp1252 vs UTF-8	Medium	Documented encoding choice
Keyword False Positives	Phrases containing keywords flagged	High	Manual review of high-risk cases required
Keyword False Negatives	Harms not containing exact keywords missed	High	Keyword list must be iteratively refined
District Column Issues	Some districts show rows=0 due to missing RecordNum	Low	Counts from flag columns remain accurate
Double-Counting	Citywide totals include district data	Medium	Use citywide file only for final totals

Case Sensitivity	All text lowercased before matching	None	Intentional - reduces missed matches

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6. Known Limitations

1. Binary Classification Only
   - Flags are True/False only
   - No severity scoring or confidence levels
   - Multiple harm types in one complaint not weighted
2. Keyword Dependency
   - Relies entirely on keyword presence in Description
   - No semantic understanding or context
   - No stemming (e.g. "leaks" ≠ "leaking")
3. Data Quality Issues
   - Assumes Description column exists and contains text
   - District column missing in some records (nan category)
   - RecordNum missing causes rows=0 display bug
4. No Temporal Analysis
   - All complaints treated equally regardless of OpenDate
   - No trending or seasonality
5. Scope Limitations
   - Only analyzes Description field
   - Ignores RecordTypeDesc, StatusCurrent, other fields
   - No cross-referencing with inspection outcomes

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7. Recommendations for Reproducibility

1. Version Control Everything
2. text

- Input CSVs (data_raw/)
- Analysis script (scripts/RUNME_LLT_ALL_HARM_AND_RISK_data_raw.py)
- Virtual environment (pip freeze > requirements.txt)
- Batch launcher (scripts/RUN_LLT_HARM_AND_RISK.bat)

3.
4. Validation Steps
5. text

1. Verify input row counts match expected
2. Spot-check 10 flagged cases per category manually
3. Confirm output CSVs have original columns + 3 new flags
4. Rerun script - outputs should be identical

6.

7. Audit Trail
    - Save console output from each run
    - Timestamp all output files
    - Document any manual interventions

Document Version: 1.0

Next Review: February 25, 2026

Status: Reproducible with documented limitations